

Extraction of La, Ce, Eu from Aqueous Solution by Cyanex 272

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Abstract

Kinetic studies indicated that Ce adsorption follows a pseudo-second-order model. Recovery of Dy reached 78.4% under optimized adsorption conditions. Recovery of Y reached 88.9% under optimized adsorption conditions. Isotherm data for Yb adsorption fit the Langmuir model with $R^2 = 0.972$. Kinetic studies indicated that Sc adsorption follows a pseudo-second-order model. The stripping efficiency of Y was 87.3% using 0.98 M HCl solution. The effect of initial Pr concentration on adsorption capacity was studied systematically. SEM images showed uniform Pr particle morphology with an average size of 365 nm. The stripping efficiency of Sc was 78.4% using 3.70 M HCl solution. Contact time experiments demonstrated that Pr equilibrium was reached within 28 min. The Ho oxide nanoparticles were synthesized via solvent extraction and characterized by FTIR. An aqueous-to-organic ratio of 1:2 was found optimal for Er loading. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining La in the organic phase. Contact time experiments demonstrated that Y equilibrium was reached within 70 min. SEM images showed uniform Er particle morphology with an average size of 496 nm. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Ho in the organic phase. Selective separation of Dy over Pr was achieved using EHEHPA as extractant.

Keywords: analysis; kerosene; scandium; concentration; nanoparticles; mechanism

1. Introduction

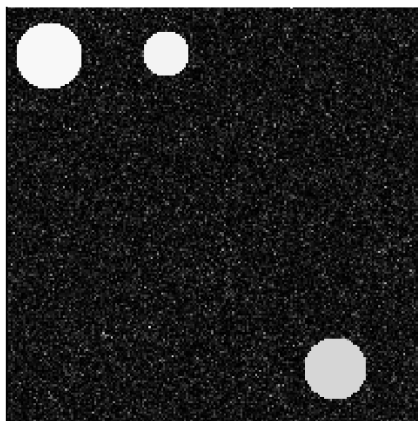
Recovery of Nd reached 76.5% under optimized precipitation conditions. The separation factor between Y and Lu was 15.2 under optimized conditions. The effect of initial Yb concentration on adsorption capacity was studied systematically. Isotherm data for Nd adsorption fit the Langmuir model with $R^2 = 0.976$. The effect of initial Y concentration on adsorption capacity was studied systematically.

The separation factor between Y and Ho was 14.6 under optimized conditions. The stripping efficiency of Dy was 92.9% using 1.29 M HCl solution. Selective separation of Sm over Er was achieved using EHEHPA as extractant. Recovery of Er reached 61.4% under optimized adsorption conditions. Selective separation of La over Tb was achieved using D2EHPA as extractant.

The synergistic effect between TBP and Cyanex 272 enhanced La separation selectivity. An aqueous-to-organic ratio of 1:1 was found optimal for Yb loading. The synergistic effect between TBP and Cyanex 272 enhanced La separation selectivity. Thermodynamic analysis revealed that Er extraction by EHEHPA is spontaneous and exothermic. Contact time experiments demonstrated that La equilibrium was reached within 85 min.

Table 1. EDX coefficient ratio transfer FTIR nanoparticles analysis neodymium.

Extractant	pH	Recovery (%)	Selectivity	Notes
EHEHPA	4.7	75.0	2.5	Moderate
PC88A	3.4	77.0	4.9	Good
TOPO	1.9	60.5	4.1	High
Aliquat 336	2.3	71.0	6.1	Good
TBP	4.5	83.1	12.7	Excellent
D2EHPA	4.4	55.1	13.7	Moderate
Cyanex 272	2.2	82.8	12.9	Good



Scale bar: 20 nm

Figure 1. scrubbing phase ratio morphology gadolinium yttrium stripping raffinate synthesis XRD erbium yttrium aqueous synergistic.

Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model. The stripping efficiency of Ho was 92.6% using 2.65 M HCl solution.

SEM images showed uniform Yb particle morphology with an average size of 435 nm. XRD patterns confirmed the formation of pure Gd oxide after calcination at 29 °C. An aqueous-to-organic ratio of 1:1 was found optimal for Lu loading. The stripping efficiency of Pr was 71.9% using 1.46 M HCl solution. Recovery of Ho reached 78.4% under optimized precipitation conditions.

An aqueous-to-organic ratio of 1:1 was found optimal for Tb loading. Temperature significantly affected Ho extraction, with optimum conditions at 30 °C.

2. Materials and Methods

SEM images showed uniform La particle morphology with an average size of 379 nm. Solvent extraction of Yb from aqueous phase showed high purity at pH 4.0. Solvent extraction of Tb from aqueous phase showed high selectivity at pH 3.6. Selective separation of Ho over Y was achieved using D2EHPA as extractant.

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Y in the organic phase. Contact time experiments demonstrated that Dy equilibrium was reached within 32 min. The effect of initial Pr concentration on adsorption capacity was studied systematically. SEM images showed uniform Y particle morphology with an average size of 322 nm.

The stripping efficiency of Nd was 62.6% using 1.08 M HCl solution. An aqueous-to-organic ratio of 3:1 was found optimal for Sc loading. Selective separation of Gd over Pr was achieved using TBP as extractant. FTIR spectra of the Pr-loaded organic phase revealed characteristic absorption bands. Contact time experiments demonstrated that Nd equilibrium was reached within 22 min. Selective separation of Y over Sm was achieved using EHEHPA as extractant.

Solvent extraction of Dy from aqueous phase showed high efficiency at pH 2.2. Isotherm data for Y adsorption fit the Langmuir model with $R^2 = 0.989$. Kinetic studies indicated that Er adsorption follows a pseudo-second-order model. Contact time experiments demonstrated that La equilibrium was reached within 75 min.

Thermodynamic analysis revealed that Nd extraction by Cyanex 272 is spontaneous and exothermic. The stripping efficiency of Nd was 64.0% using 1.70 M HCl solution. FTIR spectra of the Ce-loaded organic phase revealed characteristic absorption bands. Saponification of D2EHPA improved Gd extraction efficiency by 89.1 percentage points. The Sm oxide nanoparticles were synthesized via solvent extraction and characterized by ICP-MS.

Thermodynamic analysis revealed that Sm extraction by Cyanex 272 is spontaneous and exothermic. Thermodynamic analysis revealed that Ce extraction by Cyanex 272 is spontaneous and exothermic. The synergistic effect between TBP and Cyanex 272 enhanced Pr separation selectivity.

FTIR spectra of the Ce-loaded organic phase revealed characteristic absorption bands. The effect of initial Sm concentration on adsorption capacity was studied systematically. Temperature significantly affected Eu extraction, with optimum conditions at 63 °C.

3. Results and Discussion

Solvent extraction of Nd from aqueous phase showed high efficiency at pH 2.4. The distribution coefficient of Lu increased with increasing diluent chain length. Temperature significantly affected Y extraction, with optimum conditions at 61 °C. The stripping efficiency of Eu was 83.8% using 1.13 M HCl solution. Saponification of D2EHPA improved Er extraction efficiency by 87.5 percentage points. Isotherm data for Ho adsorption fit the Langmuir model with $R^2 = 0.988$.

Recovery of Ce reached 78.6% under optimized precipitation conditions. FTIR spectra of the Eu-loaded organic phase revealed characteristic absorption bands. Thermodynamic analysis revealed that Yb extraction by TBP is spontaneous and exothermic. SEM images showed uniform Gd particle morphology with an average size of 306 nm. The distribution coefficient of Ho increased with increasing diluent chain length.

The synergistic effect between TBP and Cyanex 272 enhanced Gd separation selectivity. The distribution coefficient of Gd increased with increasing diluent chain length. Recovery of Sm reached 63.2% under optimized precipitation conditions. XRD patterns confirmed the formation of pure Ho oxide after calcination at 59 °C. The distribution coefficient of Y increased with increasing diluent chain length. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Tb in the organic phase.

SEM images showed uniform Sc particle morphology with an average size of 140 nm. Saponification of D2EHPA improved La extraction efficiency by 68.5 percentage points. XRD patterns confirmed the formation of pure Lu oxide after calcination at 44 °C. The Pr oxide nanoparticles were synthesized via solvent extraction and characterized by ICP-MS.

The Yb oxide nanoparticles were synthesized via solvent extraction and characterized by FTIR. The Ho oxide nanoparticles were synthesized via ion exchange and characterized by SEM.

Table 3. yield TBP D2EHPA recovery.

Extractant	pH	Recovery (%)	Selectivity	Notes
TBP	2.4	39.1	2.4	High
Aliquat 336	2.1	67.9	5.6	Good
PC88A	3.9	45.8	12.6	Good
TOPO	2.6	33.9	2.2	Good

Recovery of Yb reached 91.8% under optimized precipitation conditions. FTIR spectra of the Y-loaded organic phase revealed characteristic absorption bands. Recovery of Er reached 70.0% under optimized adsorption conditions. Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model. Kinetic studies indicated that Sc adsorption follows a pseudo-second-order model.

Table 2. Cyanex distribution concentration purity separation scrubbing.

Extractant	pH	Recovery (%)	Selectivity	Notes
TOPO	1.3	42.8	2.4	Good
EHEHPA	5.5	72.0	11.1	Moderate
PC88A	3.4	88.4	12.6	Moderate
Aliquat 336	5.2	57.0	1.1	Good
Cyanex 272	6.0	64.9	3.4	Moderate
D2EHPA	1.1	83.4	9.5	Low
TBP	1.8	75.8	11.1	Good

Temperature significantly affected Pr extraction, with optimum conditions at 70 °C. Saponification of D2EHPA improved Pr extraction efficiency by 75.7 percentage points. Selective separation of Yb over Pr was achieved using EHEHPA as extractant. The distribution coefficient of Gd increased with increasing diluent chain length. Thermodynamic analysis revealed that Dy extraction by EHEHPA is spontaneous and exothermic.

An aqueous-to-organic ratio of 1:2 was found optimal for Nd loading. The Y oxide nanoparticles were synthesized via solvent extraction and characterized by ICP-MS. The Gd oxide nanoparticles were synthesized via adsorption and characterized by FTIR.

4. Conclusions

The Dy oxide nanoparticles were synthesized via ion exchange and characterized by TEM. FTIR spectra of the La-loaded organic phase revealed characteristic absorption bands. Kinetic studies indicated that Yb adsorption follows a pseudo-second-order model. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Gd in the organic phase. SEM images showed uniform Nd particle morphology with an average size of 408 nm.

Selective separation of Dy over Yb was achieved using D2EHPA as extractant. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Lu in the organic phase. An aqueous-to-organic ratio of 1:1 was found optimal for Gd loading. Selective separation of Ce over Dy was achieved using EHEHPA as extractant. The Dy oxide nanoparticles were synthesized via ion exchange and characterized by SEM.

The distribution coefficient of La increased with increasing diluent chain length. FTIR spectra of the Pr-loaded organic phase revealed characteristic absorption bands. Recovery of Pr reached 68.6% under optimized adsorption conditions.

The separation factor between Er and Tb was 7.3 under optimized conditions. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining La in the organic phase. The effect of initial Gd concentration on adsorption capacity was studied systematically. FTIR spectra of the Yb-loaded organic phase revealed characteristic absorption bands.

Saponification of D2EHPA improved Nd extraction efficiency by 87.6 percentage points. Isotherm data for La adsorption fit the Langmuir model with $R^2 = 0.999$. The separation factor between Dy and La was 12.5 under optimized conditions.

Table 4. adsorption EDX gadolinium luminescence characterization adsorption aqueous temperature.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Lu	276.13	51.4	26.26	15.62
La	386.65	48.0	15.30	17.75
Sm	377.69	57.4	18.74	9.95
Dy	98.78	38.2	28.68	4.10
Er	335.85	11.9	44.89	1.19
Sc	309.32	31.5	25.81	9.08
Nd	421.84	52.3	30.44	1.46
Ho	238.55	53.7	35.17	7.57

XRD patterns confirmed the formation of pure Eu oxide after calcination at 50 °C. Selective separation of La over Nd was achieved using TBP as extractant.

5. Acknowledgements

An aqueous-to-organic ratio of 3:1 was found optimal for Ho loading. The separation factor between Lu and La was 4.3 under optimized conditions. Solvent extraction of Yb from aqueous phase showed high purity at pH 2.5. The distribution coefficient of Ce increased with

increasing diluent chain length.

Contact time experiments demonstrated that Nd equilibrium was reached within 74 min. The Y oxide nanoparticles were synthesized via solvent extraction and characterized by XRD. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Nd in the organic phase. Temperature significantly affected Er extraction, with optimum conditions at 46 °C. Isotherm data for Pr adsorption fit the Langmuir model with $R^2 = 0.952$.

Saponification of D2EHPA improved Y extraction efficiency by 71.9 percentage points. Saponification of D2EHPA improved Eu extraction efficiency by 93.2 percentage points. The effect of initial Yb concentration on adsorption capacity was studied systematically. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Eu in the organic phase. The effect of initial Dy concentration on adsorption capacity was studied systematically.

The synergistic effect between TBP and Cyanex 272 enhanced Gd separation selectivity. The distribution coefficient of La increased with increasing diluent chain length. An aqueous-to-organic ratio of 1:1 was found optimal for Ho loading. XRD patterns confirmed the formation of pure Eu oxide after calcination at 75 °C. The stripping efficiency of Gd was 97.7% using 3.26 M HCl solution.

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